Being an isomorphism is a mere proposition, so isomorphism types of a precategory are sets

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'ct:isoprop' (source lines 95-98, Lemma 9.1.3); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 9, section 1, theorem offset 2) make the counted lemma print as Lemma 9.1.3, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: being an isomorphism is a mere proposition (Lemma 9.1.3).

Allowed standard prerequisites (editorial).

- Isomorphism in a precategory. An isomorphism from a to b is a quadruple (f, g, eta, epsilon) with eta : 1_a = g o f and epsilon : f o g = 1_b.. It is the type whose propositionality is asserted, and the two witnesses compared in the proof are such quadruples.
- Hom-sets are sets. In a precategory the types hom_A(a,b) are sets.. It is what makes the identity types of hom-sets mere propositions, so that only the first components need comparing at source line 102.
- Precategory axioms. Composition is associative and unital.. The displayed chain at source line 104 uses the unit and associativity laws to identify g with g'.

Lemma 9.1.3. *For any $f : \hom_A(a, b)$, the type "f is an isomorphism" is a mere proposition. Therefore, for any $a, b : A$ the type $a \cong b$ is a set.*

Proof. Suppose given $g : \hom_A(b, a)$ and $\eta : (1_a = g \circ f)$ and $\epsilon : (f \circ g = 1_b)$, and similarly g', η', and ϵ'. We must show $(g, \eta, \epsilon) = (g', \eta', \epsilon')$. But since all hom-sets are sets, their identity types are mere propositions, so it suffices to show $g = g'$. For this we have

$$g' = 1_a \circ g' = (g \circ f) \circ g' = g \circ (f \circ g') = g \circ 1_b = g$$

using η and ϵ'. □